MIL-C-63186(AR)
25 June 1982
Superseding
(See 6.5)

MILITARY SPECIFICATION

C-RING, METAL
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for metal C-rings used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Bags, Greaseproof, Waterproof, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-G-45204	Gold Plating, Electro-deposited

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1110

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SPECIFICATIONS (cont)

MILITARY

MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9328354	Metal C-Ring

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B-567	Measurement of Coating Thickness by the Beta Backscatter Method

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The metal C-Ring shall comply with all requirements specified on drawing (dwg.) 9328354, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

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3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

3.3 Weld quality. Weld joints of unplated welded C-Rings shall be free of cracks, pin holes, porosity or any other weld defects likely to affect the sealing characteristics of the C-Ring, when examined under 10X magnification.

3.4 Plating. The metal C-Ring jacket shall be gold plated in accordance with dwg. 9328354 and the applicable requirements of MIL-G-45204 (Type I, Grade A).

3.4.1 Plating quality. Gold plating shall be of high quality, fine-grained and free of cracks, nodules, bare spots, scratches or other defects exposing base material. In addition, there shall be no evidence of plating separation, flaking, blistering or other similar plating defects or any evidence of non-circumferential flaws or local discontinuities on the critical seating surfaces under 30X magnification. Testing shall be performed in accordance with 4.5.1.1.

3.4.2 Plating thickness. Gold plating thickness shall be as specified on dwg. 9328354. Testing shall be performed in accordance with 4.5.1.2.

3.4.3 Adhesion. Gold plating shall exhibit no evidence of defective adhesion such as separation from base material, or cracking at high temperature. Testing shall be performed in accordance with 4.5.1.3.

3.5 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.6 Workmanship. The metal C-ring shall be free of nicks, dents, burrs, or any other obvious physical defects. Parts shall be clean and free of dirt, grease or any other foreign matter. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the

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contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.5), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

Part Description	Drawings	Quantity
C-Rings (unplated)	9328354	2
C-Rings (plated)	9328354	6
Gold plating specimens (for plating tests of 4.4.3.1)	9328354	2

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspection specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the source as will be used in regular production.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Metal C-Ring	1 OF 1	9328354
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of defects and tests.			Note 1	Note 1
	Unplated C-Rings	2			
	Dimensions (unplated) Weld quantity and quality				
	Plated C-Rings	6			
	Spring assembly characteristics Dimensions (plated) Gold plating quality and surface finish Workmanship				
	Gold Plating Specimens	2			
	Gold plating thickness and adhesion				

NOTES: 1. See applicable Requirement paragraph and Paragraph Reference/Inspection Method columns of 4.4.2.1.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition a plating inspection lot shall not exceed 250 units.

4.4.2 Examinations. See MIL-A-48078 and the quality conformance inspection tables herein.

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QUALITY CONFORMANCE INSPECTION

MIL-C-63186(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Metal C-Ring (see note 1)	1 OF 1	9328354
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter		100%	3.1	Gage
102.	.095 width of jacket (2 sides)		100%	3.1	Gage
103.	.010 jacket thickness (see note 2)		-	3.1	Gage
104.	Weld quantity and quality (Welded C-Rings only)		100%	3.1, 3.3	Visual/10X
105.	Gold plating thickness and adhesion	4.4.3.1		3.4.2, 3.4.3	4.5.1.2, 4.5.1.3
106.	Spring improper, missing or improperly assembled after plating		100%	3.1	Visual
107.	.125 free height over spring after plating		100%	3.1	Gage
108.	.113 max. height over spring after plating		100%	3.1	Gage
109.	.060 flatness after plating		100%	3.1	Gage
110.	Gold plating quality after plating		100%	3.4.1	4.5.1.1
111.	Surface finish after plating		100%	3.1	Visual
112.	Poor workmanship		100%	3.6	Visual
113.	Materials		-	3.2	Visual (6.4)

NOTES:
1. Characteristics are before plating unless otherwise specified
2. Inspect on material before fabrication

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63186(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		1 Pk./ Lot	5.0	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		-	5.1.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1.1	Visual
104.	Parts not individually packaged		100%	5.1.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package		100%	5.1.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1.2	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1.3	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1.4	Visual
110.	Evidence of poor workmanship		100%	-	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Plating thickness and adhesion. Sample specimens from each plating inspection lot plated concurrently with the actual items shall be tested for these requirements in accordance with 4.5.1.2 and 4.5.1.3. The specimens shall be prepared from the same base material in essentially the same configuration as the actual items, and cleaned and processed concurrently with the items to be plated. The test specimens shall be at least two in number and located in positions which will register quality representative of the plating process. If any specimen fails to meet any of the applicable requirements, the plating inspection lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test Methods and Procedures.

4.5.1 Plating (see 3.4).

4.5.1.1 Quality. Gold plating quality of the external surfaces of the C-Ring shall be examined on each item under 30X magnification with special attention to assuring that there are no plating flaws or non-circumferential scratches on the critical seating areas specified on the dwg.

4.5.1.2 Thickness. Gold plating thickness of specimens shall be measured in accordance with ASTM B-567 or other approved test method.

4.5.1.3 Adhesion. Gold plating adhesion of specimens shall be examined in accordance with 4.5.2.3 of MIL-G-45204 by heating samples at 250° to 300°F for a minimum of one hour, cooling and examining under 4X magnification for any evidence of flaking, peeling, blistering or any other plating separation from the base material.

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5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be single item packaged in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from MIL-B-121 (Type II, Grade A, Class 1) material or equivalent. Fillers, positioners, stiffeners and cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate packing. Single item packages shall be packed in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packs and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturer
e. Contract or purchase order no.
f. Lot number
g. Date of acceptance

6. NOTES

6.1 Intended use. Metal C-Rings covered by this specification are intended for use on rocket motor/bulkhead assemblies for XM753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.5).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of AR-PD-11 dated 9 April 1979 with Amendment 1 dated 23 December 1980.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

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Preparing activity

Army-AR

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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